INTERNSHIP PROJECT REPORT

UART TRANSMITTER AND RECEIVER DESIGN USING VERILOG HDL

Submitted by

B MANJUSHA (410623106010)

In partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

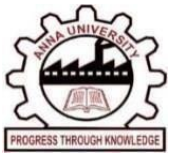
In

ELECTRONICS AND COMMUNICATION ENGINEERING



DHAANISH CHENNAI COLLEGE OF ENGINEERING

PADAPPAI, CHENNAI - 601 301



ANNA UNIVERSITY, CHENNAI - 600 025

OCT 2026

Abstract

This project implements a simplified UART (Universal Asynchronous Receiver/Transmitter) transmitter and receiver using Verilog HDL. The transmitter sends an 8-bit data word serially, with a start bit and a stop bit. The receiver detects the start condition, samples the serial input in the simulation model, reconstructs the 8-bit word, and raises a completion signal. The design was tested using a Verilog testbench and simulated with Icarus Verilog. For the test byte 0xA5, the observed result was “Received data = a5” and “Done = 1”. This is an educational synchronous simulation model, not a complete physical UART implementation with baud-rate generation or mid-bit sampling.

1. Introduction

UART is a serial communication method commonly used to transfer data between digital systems. A transmitter converts parallel data into a serial bit stream, while a receiver collects serial bits and reconstructs the original byte. A typical UART frame contains an idle-high line, a low start bit, data bits transmitted least-significant-bit first, and a high stop bit.

2. Objectives

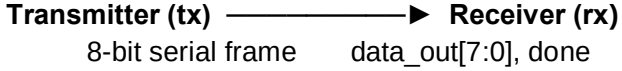
- Design a basic 8-bit serial transmitter using Verilog HDL.
- Design a receiver that reconstructs the transmitted byte.
- Connect the transmitter output to the receiver input in a testbench.
- Verify the design through simulation and observe the completion signal.
- Understand state-machine-based digital design and waveform-based debugging.

3. Tools and Technologies

- Hardware Description Language: Verilog HDL
- Software Tool: VeriSim
- Simulator: Icarus Verilog (iverilog and vvp)
- Waveform output: VCD (Value Change Dump), viewed in a waveform viewer
- Design files: design.v and testbench.v

4. System Overview

The testbench instantiates the transmitter module (example) and receiver module (uart_receiver). The transmitter’s tx output is connected directly to the receiver’s rx input. Both modules use the same simulation clock. The transmitter sends the fixed test byte 8'hA5 when start is asserted. The receiver stores the incoming bits and updates data_out when the frame is accepted. The done output indicates completion for one clock event in this model.



5. Simulation Procedure

1. Save the transmitter and receiver modules in design.v.
2. Save the testbench in testbench.v.
3. Compile using: iverilog -g2005 -o out.vvp design.v testbench.v
4. Run using: vvp out.vvp
5. Inspect the console output and the VCD waveform.

6. Results and Discussion

Observed console output:

```
Received data = a5
Done = 1
```

The received byte 0xA5 matches the test byte sent by the transmitter, and done was asserted to indicate reception completion. The waveform was also checked for tx, data_out, and done.

7. Conclusion

The project demonstrated a basic serial data transfer path using Verilog HDL. The transmitter sent the test byte 0xA5, and the receiver reconstructed the same byte in simulation. The project provided practical experience with finite-state machines, serial data handling, testbench development, simulation, and waveform inspection.